Evaluating the Deployment Utility of a Frozen Policy-Update Procedure: Scale, Completion Preference, and Joint Evidence

Research Presentation

Scope of This Study

Empirical and methodological evaluation of one frozen replay-proxy policy-update procedure in a controlled synthetic continuous double auction, using saved outcomes, root-level inference, and explicit distinctions between confirmatory and post-outcome diagnostics.

Why Proxy Scores Are Insufficient for Deployment Assessment

- Candidate policy updates are frequently screened on cheap replay or proxy benchmarks before interactive deployment.
- A positive proxy gain does not guarantee positive incumbent-relative utility once the selected policy interacts with an active environment.
- Four distinct mechanisms can separate proxy screening from deployed utility:
  1. **Winner selection:** choosing among multiple candidates induces selection optimism on finite proxy samples.
  2. **Interactive feedback:** deployed orders alter book state, liquidity, and subsequent fills unlike passive replay.
  3. **Operating scale:** order size changes market impact and execution completion rates.
  4. **Completion preferences:** deployment utility depends on how incomplete fills are penalized relative to execution prices.
- Sign-reversal frequency alone is insufficient: deployment value depends on the magnitudes of gains and losses, including no-update decisions.

Procedure-Level Estimand and Research Questions

Formal Target (Procedure-Level Deployment Utility)

Expected incumbent-relative utility of the *complete frozen procedure*—retaining both candidate adoption and no-update decisions—under declared operating conditions, candidate-generation population, utility function, and joint evidence family.

What this paper does not propose:

- No new off-policy estimator, fidelity metric, selection algorithm, or universal deployment theorem.

Primary Research Questions:

1. **Operating scale & liquidity:** How does deployed incumbent-relative utility vary across order scales ($s \in \{0.002, 0.05, 0.3\}$), liquidity regimes (liquid, thin), and inventory responses ($\rho \in \{0, 1\}$)?
2. **Utility and evidence sensitivity:** How does fixed-selection revaluation across incomplete-fill penalties ($p \in \{25, 50, 100, 200\}$ bps), together with the choice of inferential family, change the deployment assessment?

Auxiliary diagnostic: a separate post-outcome proxy-to-deployment comparison is reported at the selection scale ($s = 0.05$); it is not a primary research question.

Related Work and Scope Boundary

Research Area	Relationship and Boundary to This Work
Off-Policy Evaluation (OPE) & High-Confidence OPE	Evaluates policies from logged data. We evaluate paired interactive deployments of a frozen replay-screened update procedure.
Safe Policy Improvement (SPIBB)	Constructs safer policy improvements under uncertainty. We audit a fixed procedure without altering its selection rule.
Conservative Evaluation of Offline Policy Learning	Close TMLR precedent on rigorous evaluation of offline learning pipelines. We focus on procedure-level deployment utility across scale and penalty preferences.
Active Offline Policy Selection	Adaptively allocates limited online evaluation budgets across candidates. Here, selection is frozen on replay proxies prior to deployment assessment.
HAL & Agent Reliability	Studies system-level benchmarking and ranking measurement. We evaluate a specific frozen market-execution update procedure.
PIVOT (Improvement Fidelity & PIVOT-KG)	Studies transition-level Improvement Fidelity and PIVOT-KG. This paper evaluates frozen procedure-level deployment utility and does not replace those methods or repeat PIVOT results.

Formal Characterization 1: Reversal Accounting

These are elementary formal characterizations, not new estimators or deep general theorems.

Gain and loss magnitudes matter

Let A denote adoption; an incumbent-retaining decision has zero procedure gain. Then

$$\mathbb{E}[\Delta U] = \Pr(A)[\Pr(\Delta U \geq 0 \mid A)\mathbb{E}[\Delta U \mid A, \Delta U \geq 0] - \Pr(\Delta U < 0 \mid A)\mathbb{E}[|\Delta U| \mid A, \Delta U < 0]] \,.$$

Interpretation

Adoption or reversal frequency alone cannot determine procedure value. Two procedures can have the same reversal rate but opposite means when their positive and negative outcome magnitudes differ.

The estimand retains no-update decisions; it is not the mean of adopted updates only.

Formal Characterizations 2–3: Utility and Sampling

2. Fixed-selection completion-utility frontier

For the same selected policies and deployments, separating the execution component a from completion shortfall b gives

$$\mu(p) = a + p\,b.$$

A mathematical zero, when $b \neq 0$, is $p^* = -a/b$. Changing p does not reselect candidates.

3. Root/deployment variance decomposition

For R independent roots and m repeated deployments per root,

$$\mathrm{Var}(\widehat{\mu}) = \frac{\mathrm{Var}_R(g_R)}{R} + \frac{\mathbb{E}_R[\sigma_R^2]}{Rm}.$$

More repetitions reduce within-root deployment noise; only more independent roots reduce root-population variation.

Why it matters here

The 2,400 paired deployment records improve within-root precision, but the primary independent unit remains the 50-root cluster.

Frozen Experimental Protocol and Statistical Units

Frozen selection and evaluation design

- **Environment:** Synthetic continuous double auction.
- **Procedure:** One frozen replay-proxy policy-update procedure.
- **Generation roots:** 50 independent generation roots; each root screens 16 candidate policies.
- **Selection decisions:** 100 root–regime selection decisions: one for each of 50 roots in each liquidity regime.
- **Interactive evaluation:** 2,400 paired interactive deployment records; selected policies and incumbents are evaluated on separate interactive books.

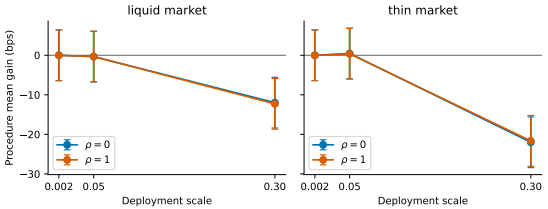
Operating grid and confirmatory family

- **Liquidity regimes:** Liquid and thin.
- **Order scales:** $s \in \{0.002,\ 0.05,\ 0.3\}$.
- **Inventory response:** $\rho \in \{0,\ 1\}$.
- **Confirmatory utility:** U_{100} with a 100-bps incomplete-fill penalty.
- **Multiplicity control:** Approximate simultaneous confidence band over a prespecified 24-estimate confirmatory family.

Independent unit of inference

The independent statistical unit is the **root** ($R = 50$), not each deployment record. Repeated deployments reduce within-root noise without adding new roots.

Main Scale Results Under U_{100}



Largest-Scale ($s = 0.3$) Cell Means (bps)

Original 24-estimate simultaneous intervals:

Cell ($s = 0.3$)	Mean	Simultaneous CI
Liquid, $\rho = 0$	-11.9800	$[-18.3944, -5.5656]$
Liquid, $\rho = 1$	-12.2848	$[-18.6992, -5.8704]$
Thin, $\rho = 0$	-22.0073	$[-28.4217, -15.5929]$
Thin, $\rho = 1$	-21.6446	$[-28.0590, -15.2302]$

- At $s = 0.3$, all four regime-by-response cells exhibit clearly negative incumbent-relative deployment utility.
- Thin liquidity doubles the magnitude of the largest-scale loss relative to liquid books.

What the Prespecified 24-Estimate Joint Family Supports

Supported by Original Joint Family

- Negative incumbent-relative deployment utility under U_{100} at the largest order scale ($s = 0.3$) in all four operating cells:
  - Liquid, $\rho = 0$: -11.9800 bps
  - Liquid, $\rho = 1$: -12.2848 bps
  - Thin, $\rho = 0$: -22.0073 bps
  - Thin, $\rho = 1$: -21.6446 bps
- Every corresponding simultaneous confidence interval excludes zero by at least 5.56 bps.

Unresolved Under Original Joint Family

- Lower-scale cell means at $s = 0.002$ and $s = 0.05$ are not resolved as strictly positive or negative under the 24-estimate simultaneous band.
- Inventory-response contrasts ($\rho = 1$ vs. $\rho = 0$) and interaction contrasts remain unresolved under the original family.
- Absence of a resolved effect in these smaller-scale or contrast cells is treated as inconclusive, not as proof of zero effect.

Post-Outcome Proxy-to-Deployment Gap Diagnostic ($s = 0.05$)

Outcome-Aware Diagnostic Specification

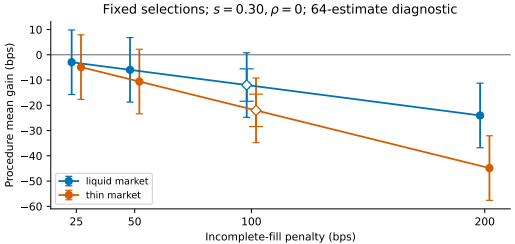
Compares saved replay-proxy gains (with no-update decisions set to zero) against paired interactive deployment gains at the selection scale $s = 0.05$, using a separate whole-root simultaneous interval construction.

Cell	Gap (bps)	Whole-root simultaneous interval
Liquid, $\rho = 0$	1.314	[0.649, 1.979]
Liquid, $\rho = 1$	1.294	[0.629, 1.959]
Thin, $\rho = 0$	0.952	[0.288, 1.617]
Thin, $\rho = 1$	1.057	[0.392, 1.722]

Explicit Methodological Caveats:

- **Post-outcome:** Separate diagnostic, outside the original 24-estimate confirmatory family.
- **Composite gap:** Combines winner selection across 16 candidates, proxy-sample noise, and replay–interactive differences.
- **No causal decomposition:** Not an isolated or causal estimate of proxy bias.

Post-Outcome Completion Preference Sensitivity and Break-Even



Post-Outcome Diagnostic Scope

- Revalues the *same frozen selected policies* at incomplete-fill penalties $p \in \{25, 50, 100, 200\}$ bps.
- Does **not** reselect candidates under alternative penalties.
- The 64-estimate expanded family, root-deletion audits, alternative simultaneous bands, and break-even analysis are all **post-outcome diagnostics** separate from the confirmatory 24-estimate family.

Affine break-even diagnostic

Uses $\mu(p) = a + p\,b$ with a separate joint component construction. It is descriptive and does **not** establish a confirmatory threshold.

Specification for an Auditable Deployment Assessment Report

To make procedure-level evaluation reproducible and transparent, an assessment report should explicitly separate frozen design choices from post-outcome sensitivity checks:

Report Component	**Required Specification in This Protocol**
1. Procedure & Population	Frozen update rule with no-update retention; 50 roots, 16 candidates per root, and 100 root–regime decisions.
2. Operating Grid	Explicit environment regimes (liquid, thin), order scales ($s \in \{0.002, 0.05, 0.3\}$), and inventory responses ($\rho \in \{0, 1\}$).
3. Utility Accounting	Primary U_{100} with a 100-bps incomplete-fill penalty; 2,400 paired interactive records.
4. Unit of Inference	Root-level clustering ($R = 50$) recognizing that repeated deployments reduce within-root variance $\sigma^2_{\text{dep}}/(Rm)$ without adding roots.
5. Multiplicity Boundary	Strict separation between the prespecified 24-estimate confirmatory family and post-outcome diagnostics (64-estimate family, proxy gap, break-even, root-deletion audits).

Limitations and Scope Boundaries

- **Single procedure and synthetic market:** All empirical evidence is conditional on one frozen replay-proxy update procedure and one controlled synthetic continuous double auction.
- **Saved-outcome reanalysis:** Current results are a reanalysis of saved outcomes rather than a full raw-model rerun.
- **No broad generalization claims:** The study makes no claim of a universal scaling law, general agent reliability, real-market execution validity, or isolated causal proxy bias.
- **Unmeasured operational dimensions:** Does not measure API monetary cost, token consumption, inference latency, safety constraints, or long-horizon reliability.
- **Auxiliary planner-code case:** The planner-code illustration uses only 3 generation roots and serves strictly as an auxiliary check, not as statistical evidence for unrestricted coding or autonomous agents.

Takeaways and Conclusion

1. **Evaluate the full frozen procedure:** Deployment utility must account for candidate screening, no-update decisions, and interactive feedback at the level of independent generation roots ($R = 50$).
2. **Scale dominates confirmatory conclusions:** Under the prespecified 24-estimate family for U_{100}, all four largest-scale cells ($s = 0.3$) are significantly negative (-11.98 to -22.01 bps), while smaller scales ($s = 0.002, 0.05$) and response contrasts remain unresolved.
3. **Proxy gains do not transfer directly:** At the selection scale ($s = 0.05$), post-outcome whole-root diagnostics show a positive gap of 0.952 to 1.314 bps between saved replay-proxy gains and interactive deployment gains.
4. **Completion preferences matter:** Post-outcome revaluation across $25, 50, 100$, and 200 bps along the affine frontier $\mu(p) = a + p\,b$ demonstrates how incomplete-fill penalties govern sign and magnitude at large order scales.
5. **Disciplined reporting:** Distinguishing prespecified joint confirmatory inference from post-outcome diagnostics prevents overclaiming from multi-condition evaluation suites.